Fermat's Theorem tells us that if a function f has a local extremum at a point c, then c must be a critical point of f. However, a function is not guaranteed to have a local extremum at a critical point.

Example. $f(x) = x^3$ has a critical point at $x = 0$, but $f(0) = 0$ is neither a local max nor a local min since $f(x) < 0$ for every $x < 0$ and $f(x) > 0$ for every $x > 0$.

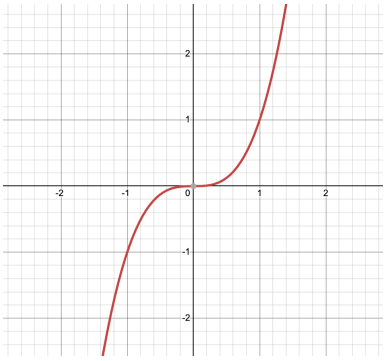


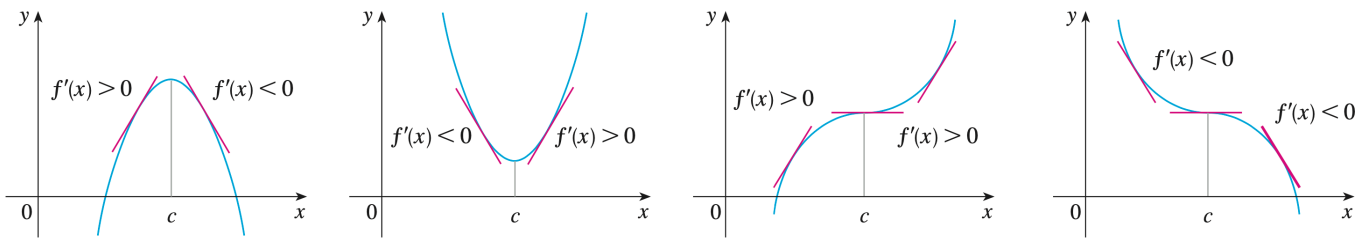
Figure 1: The function x^3 has neither a local max nor a local min at the critical point $x = 0$.

Since the locations of relative extrema are key information in sketching the graph of f, it is therefore desirable to be able to classify the critical points of a given function. The next theorem provides such a classification for a large class of functions.

Theorem. (First Derivative Test) Suppose that f is a continuous function over an interval I containing a critical point c. If f is differentiable over I, except possibly at point c, then exactly one of the following is true about $f(c)$:

- If $f'(x) > 0$ when $x < c$ and $f'(x) < 0$ when $x > c$, then $f(c)$ is a local maximum.
- If $f'(x) < 0$ when $x < c$ and $f'(x) > 0$ when $x > c$, then $f(c)$ is a local minimum.
- If $f'(x)$ has the same sign for $x < c$ and $x > c$, then $f(c)$ is neithr a local maximum nor a local minimum.

It is easy to remember the First Derivative Test by visualizing diagrams such as those below:



(a) Local maximum (b) Local minimum (c) No maximum or minimum (d) No maximum or minimum

1. Explain why the First Derivative Test follows from Corollary 3 of the Mean Value Theorem.

(The rest of this page is optional.) To prove the First Derivative Test, we will need the following lemma.

Lemma.

- If f is increasing (resp., decreasing) on the interval (a,c) and is continuous on $(a,c]$, then f is increasing (resp., decreasing) on $(a,c]$.
- If f is increasing (resp., decreasing) on the interval (c,b) and is continuous on $[c,b)$, then f is increasing (resp., decreasing) on $[c,b)$.

Proof. (of Lemma) Suppose f is increasing on (a,c) and continuous on $(a,c]$. Then since f is increasing on (a,c) for any $x \in (a,c)$ and $y \in (x,c)$ we have $f(x) < f(y)$, so it follows that $\lim_{y\to c^+} f(x) \leq \lim_{y\to c^+} f(y)$ and therefore $f(x) \leq f(c)$. This shows that f is increasing on $(a,c]$. The proofs of the remaining cases are similar and are left as an exercise for the interested reader. □

Proof. (of First Derivative Test)

- Suppose f is increasing on (a,c) and decreasing on (c,b). Since f is continuous at c, by the Lemma f is increasing on $(a,c]$ and decreasing on $[c,b)$. If $x \in (a,c)$, then since f is increasing on $(a,c]$ we have $f(x) \leq f(c)$. If $x \in (c,b)$, then since f is decreasing on $[c,b)$ we have $f(x) \leq f(c)$. This shows that $f(x) \leq f(c)$ for all $x \in (a,b)$, hence $f(c)$ is a local maximum.
- The proof of the case when f changes from decreasing to increasing at c is similar and left as an exercise for the interested reader.
- Suppose f is increasing on (a,c) and increasing on (c,b). Then by the Lemma f is increasing on $(a,c]$ and increasing on $[c,b)$. As shown above $f(x) \leq f(c)$ for all $x \in (a,c)$. If $x \in (c,b)$, then since f is increasing on $[c,b)$ we have $f(c) \leq f(x)$ for all $x \in (c,b)$. Thus there is no open interval containing c for which $f(x) \leq f(c)$ for all points in the open interval, since $f(c) \leq f(x)$ for any point x to the right of c. Hence, $f(c)$ is not a local maximum. Similarly, $f(c)$ is not a local minimum. The proof of the case where f is decreasing to the left and right of c is similar and left as an exercise for the interested reader.

□

2. Find all local extrema of the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$. Confirm your answer by checking the graph in Desmos.

3. Find all local extrema of the function $f(x) = 5x^{1/3} - x^{5/3}$. Confirm your answer by checking the graph in Desmos.